Find the k-th Smallest Pair Distance

Let $a := \langle a_1, \ldots, a_n \rangle$ be a sequence of n integers. The *distance* of an ordered pair (i, j) with $1 \leq i < j \leq n$ is defined as $|a_i - a_j|$.

Let $d := \langle d_1, \ldots, d_m \rangle$ be the sequence of distances of all $m = \frac{n(n-1)}{2}$ pairs, in increasing order (if $s \leq t$, then $d_s \leq d_t$). Compute d_k, the k-th smallest distance.

Example Test Cases:

Test Case 1
Input: $n = 3$, `a` = `[1,3,1]`, $k = 1$
Output: 0
Explanation: The distances of all pairs are:
- $|a_1 - a_2| = |1 - 3| = 2$
- $|a_1 - a_3| = |1 - 1| = 0$
- $|a_2 - a_3| = |3 - 1| = 2$

The sequence d is $\langle \underline{0}, 2, 2 \rangle$. The 1st smallest distance is 0.

Test Case 2
Input: $n = 4$, `a` = `[1,6,1,5]`, $k = 3$
Output: 4
Explanation: The sequence d is $\langle 0, 1, \underline{4}, 4, 5, 5 \rangle$. The 3rd smallest distance is 4.

Expected Time Complexity: $O(n \log n + n \log W)$
(where $W = \max(a) - \min(a)$)

Input Format:
An integer n, an array `a` of length n, and an integer k.

Output Format:
Return a single integer, the k-th smallest distance, as defined above.

Constraints:

1. $2 \leq n \leq 10^6$
2. $0 \leq a_i \leq 10^9$
3. $1 \leq k \leq \dfrac{n(n-1)}{2}$

Instructions:

You have been provided with two template files, containing the `main` and `solve` functions respectively. The `main` function, which has been completed for you, parses the input for each test case. For each test case, it calls the `solve` function, which must compute the k-th smallest distance.

The `solve` function must compute this in $O(n \log n + n \log W)$ time.

(Refer to the `Instructions.pdf` file for detailed guidelines on the input-output format and additional test cases.)

Library Usage

For this lab, you are permitted to use any modules which are part of the standard library of either `C++` or `python`. For `C++`, do not add any extra `#include` directives; the provided `bits/stdc++.h` covers all allowed modules.